

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (CL) Examination
May 2013

CL 309 Construction Planning and Management

Date: 03.05.2013

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) 'Project Management is tool for economy in construction industries'. Justify the statement in context with benefits of project management. 04
- (b) State advantages and disadvantages of Bar Chart method. 03
- Q - 2 (a) Construct an activity network for a project for following details. 08

Activity	Time in weeks	Activity	Time in weeks
1-2	4	5-6	4
1-3	1	5-7	8
2-4	1	6-8	1
3-4	1	7-8	2
3-5	6	8-10	5
4-9	5	9-10	7

- i. Indicate Critical path.
 - ii. Calculate expected time and variance.
- (b) Define: Direct cost, Indirect cost, Critical path, Crash cost, Cost-slope, Dummy activity. 06

OR

- Q - 2 (a) A project consist of the activity shown in table below: 10

Activity	A	B	C	D	E	F	G	H
Immediate Predecessors	-	-	-	-	B	C	D	E,G
Time in Days	4	2	2	4	3	2	3	4
Manpower needed every day	9	3	6	4	8	7	2	1

- i. Draw the project network.
- ii. Find critical activities and normal project duration.
- iii. Find schedule which minimize the day today variation in manpower requirement.(resource leveling)

- (b) How the probability of completion time for a project can be determine? Discuss steps involve for it. 04

Q - 3 (a) Why Critical path method is suitable for repetitive type of projects? State salient features of this technique. 04

- (b) Compute early start, early finish, latest start and latest finish time for activities of following network in tabular format. 05

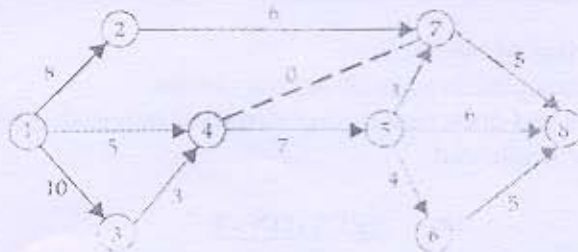


Figure 1.

- (c) Write short notes on Depreciation 05

OR

Q - 3 (a) With relevant example define Earliest event time and Latest event time. Explain its use in critical path method. 03

- (b) The network for a project with estimated duration in days is shown in Figure 2. Determine the total float for each activity and establish the critical path. 06

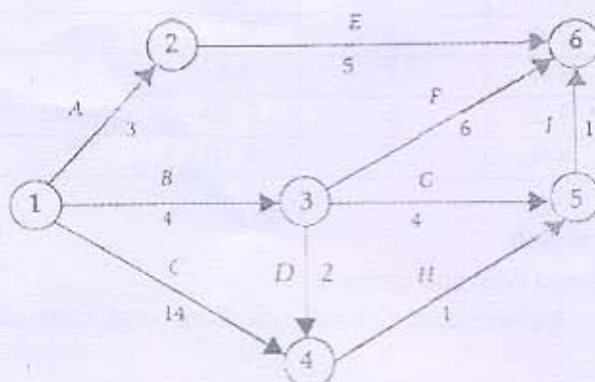


Figure 2.

- (c) A construction firm is considering the following two alternatives: 05

- Lease a new road roller for ₹ 85,000 per year, payable in advance. The system should last 6 yr.
- Rent at ₹ 90,000 after end of every year.

Which alternative should be selected using a discounted present worth analysis? If the MARR is 8%.

SECTION - II

Q - 4 (a) Explain the functions of construction management 07

Q - 5 (a) What is monitoring? Why monitoring is importance of it in project management. 04

- (b) What is organization? Explain the types of organization. 05

(c) State various phases of project management. 05

OR

Q – 5 (a) Which are the temporary services generally required on construction site. Explain it. 04

(b) State processes of controlling and explain project controlling techniques. 05

(c) What are the causes of failure in construction projects? 05

Q – 6 (a) Explain the effect of Slope, Altitude and temperature on performance of I.C. engine of construction equipment. 06

(b) Write short note on scraper. 06

(c) State the characteristics of Indian labours. 02

OR

Q – 6 (a) Write short note on compacting equipments. Discuss how soil parameters govern the selection of compacting equipment. 06

(b) 'Hauling equipments are important in construction industry.' Justify the statement. 06
Also state types and factors affecting production of hauling equipments.

(c) State the factors affecting efficiency of Indian labour. 02
